Chapter 2. Identification and Adjustment

Interventions and the truncated factorization, confounding, the back-door and adjustment criteria

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2.1 Identification of causal effects

Motivation. How an SCM defines a population under intervention, and when that distribution follows from the diagram and observational data.

2.1.1 Structural causal models, revisited

Motivation. Recall the framework and introduce the do(·) operator.

2.1.1.1 Observation versus intervention

We learn about reality by *observation* (passive records) and *intervention* (trials, experiments). Questions on drug efficacy or hypothetical policies concern the system *under intervention*, even when only observational data exist.

An SCM $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}) \rangle$ models the unknown reality: measured $\mathbf{V}$, unknown $\mathbf{U}$ (what each unit is), and $v \leftarrow f(\cdot)$ for each V. It induces a diagram and a family of distributions:

$$\begin{aligned} \mathcal{M} &\longmapsto \mathcal{G}, \\ \mathcal{M} &\longmapsto \{P,\ P_{x_1},\ P_{x_2},\ P_{x_2,x_4}, \dots\} \end{aligned}$$

for $\text{do}(\emptyset), \text{do}(x_1), \text{do}(x_2), \text{do}(x_2, x_4), \dots$

Layer 1 (observational). As in Chapter 1,

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{v}, \mathbf{u}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid v_1, \dots, v_{i-1}, \mathbf{u}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i);$$

Layers 2 and 3 are induced as well.

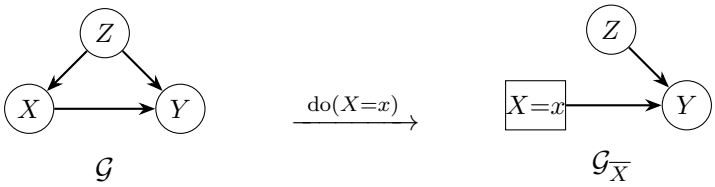
Causal diagram. $B \to V_i$ if B is an argument of f_i; $V_i \leftrightarrow V_j$ if $\mathbf{U}_i, \mathbf{U}_j$ correlate.

2.1.1.2 The do-operator

Definition (Intervention)

Given $\mathcal{M}$, fixing $X \in \mathbf{V}$ to a constant x is written $\text{do}(X = x)$. It transforms $\mathcal{M}$ into the *submodel* $\mathcal{M}_x$, i.e. $\mathcal{M}$ with $f_X \in \mathcal{F}$ replaced by the constant x.

The diagram of $\mathcal{M}_x$ is $\mathcal{G}_{\overline{X}}$: edges *into* X removed.



$\mathcal{G}$ $\qquad$ $\mathcal{G}_{\overline{X}}$

2.1.2 The second layer: causal effects

Motivation. Define causal effects and compute them in Markovian models.

2.1.2.1 Connecting different worlds

Data come from P, the query from P': would banning smoking reduce cancer? $Q = P(\text{Cancer} = \text{true} \mid \text{do}(\text{Smoking} = \text{no}))$ is *not* an aspect of P.

Observation 1. A distribution describes a population under one regime, not change.
Observation 2. We must represent how the population reacts to a change of regime.

The challenge. Age/sex Z, action X, mediator W, outcome Y: $Z \to X \to W \to Y$, $Z \to Y$. Setting $X = x$ cuts $Z \to X$; reality gives $P(z, x, w, y)$, the hypothetical world $P(z, w, y \mid \text{do}(x))$. *Causal effects* are changes of Y when X is varied between constants, free of Z. Is $P(y \mid \text{do}(x)) = P(y \mid x)$?

Definition (Causal effect; Pearl 2000, Def. 3.2.1)

Given disjoint sets $\mathbf{X}, \mathbf{Y}$, the causal effect of $\mathbf{X}$ on $\mathbf{Y}$, $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$, is a function from $\mathfrak{X}_{\mathbf{X}}$ to the space of distributions over $\mathbf{Y}$. For each $\mathbf{x}$, $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$ is the probability of $\mathbf{Y} = \mathbf{y}$ induced by deleting from the model all equations of the variables in $\mathbf{X}$ and substituting $\mathbf{X} = \mathbf{x}$ in the remaining equations:

$$P(\mathbf{y} \mid \text{do}(\mathbf{x})) = P_{\mathbf{x}}(\mathbf{y}), \quad \text{the probability of } \mathbf{Y} = \mathbf{y} \text{ in } \mathcal{M}_{\mathbf{x}}.$$

2.1.2.2 Computing causal effects from observational data (Markovian case)

In the example,

$$P(\mathbf{v}) = P(z)\,P(x \mid z)\,P(w \mid x)\,P(y \mid w, z),$$
$$P_x(\mathbf{v} \setminus \{X\}) = P(z) \cdot \underbrace{1}_{X=x \text{ in } \mathcal{M}_x} \cdot P(w \mid x)\,P(y \mid w, z),$$

the other mechanisms being untouched (*modularity*). Hence

$$P(y \mid \text{do}(x)) = \sum_{z,w} P(z)P(w \mid x)P(y \mid w, z) \;\neq\; \sum_{z,w} P(z \mid x)P(w \mid x)P(y \mid w, z) = P(y \mid x)$$

in general: conditioning weights z by $P(z \mid x)$, intervening by $P(z)$.

Example 2.1 (Sprinkler: seeing versus doing)

Season $\to$ Sprinkler, Season $\to$ Rain, Sprinkler $\to$ Wet $\leftarrow$ Rain, Wet $\to$ Slippery.

$$Q_1 = P(\text{wt} \mid \text{Sp} = \text{on}) = \frac{\sum_{\text{sn,rn}} P(\text{sn})P(\text{Sp} = \text{on} \mid \text{sn})P(\text{rn} \mid \text{sn})P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn})}{\sum_{\text{sn}} P(\text{Sp} = \text{on} \mid \text{sn})P(\text{sn})},$$
$$Q_2 = P(\text{wt} \mid \text{do}(\text{Sp} = \text{on})) = \frac{\sum_{\text{sn,rn}} P(\text{sn})\,P'(\text{Sp} = \text{on})\,P(\text{rn} \mid \text{sn})P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn})}{\sum_{\text{sn}} P'(\text{Sp} = \text{on})P(\text{sn})}$$
$$= \sum_{\text{sn,rn}} P(\text{sn})P(\text{rn} \mid \text{sn})P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn}),$$

with $P'(\text{Sp} = \text{on}) = 1$. Seeing the sprinkler on is evidence about season and rain; turning it on is not.

2.1.3 Truncated factorization

Motivation. The general interventional formula for Markovian models, read as reweighting.

Theorem 2.1 (Truncated factorization; manipulation theorem, g-computation)

In a Markovian model, the distribution generated by $\mathrm{do}(\mathbf{X}=\mathbf{x})$ is

$$P(\mathbf{v} \mid \mathrm{do}(\mathbf{x})) = \begin{cases} \prod_{V_i \in \mathbf{V}\setminus\mathbf{X}} P(v_i \mid \mathbf{pa}_i) & \text{if } \mathbf{v} \text{ is consistent with } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases}$$

Derivation [Supplement]. $\mathcal{M}_{\mathbf{x}}$ is Markovian with the same f_i, $P(u_i)$ for $V_i \notin \mathbf{X}$ and constant mechanisms for $\mathbf{X}$; its Bayesian factorization (Chapter 1) is

$$\prod_{V_i \notin \mathbf{X}} P_{\mathbf{x}}(v_i \mid \mathbf{pa}_i) \prod_{X \in \mathbf{X}} \mathbb{1}[x \text{ agrees with } \mathbf{x}].$$

As f_i, U_i are unchanged and $U_i \perp\!\!\!\perp Pa_i$ in both models,

$$P_{\mathbf{x}}(v_i \mid \mathbf{pa}_i) = \sum_{u_i} \mathbb{1}[f_i(\mathbf{pa}_i, u_i) = v_i] P(u_i) = P(v_i \mid \mathbf{pa}_i)$$

whenever defined (positivity). ■

Reweighting form. Since $P(\mathbf{v}) = \prod_{V_i \in \mathbf{V}\setminus\mathbf{X}} P(v_i \mid \mathbf{pa}_i) \cdot \prod_{X \in \mathbf{X}} P(x \mid \mathbf{pa}_X)$,

$$P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x})) = \frac{P(\mathbf{v})}{\prod_{X \in \mathbf{X}} P(x \mid \mathbf{pa}_X)}, \qquad \mathbf{v}' = \mathbf{v} \setminus \mathbf{x}.$$

For singleton X, with $\mathbf{v}'' = \mathbf{v} \setminus (\mathbf{pa}_X \cup \{x\})$,

$$P(\mathbf{v}' \mid \mathrm{do}(x)) = \frac{P(\mathbf{v})}{P(x \mid \mathbf{pa}_X)} = \frac{P(\mathbf{v}) P(\mathbf{pa}_X)}{P(x, \mathbf{pa}_X)} = P(\mathbf{v}'' \mid x, \mathbf{pa}_X)\, P(\mathbf{pa}_X).$$

Units with $X = x$ are up-weighted by $1/P(x \mid \mathbf{pa}_X)$ (origin of IPW, Chapter 6).

2.1.4 The identification problem

Motivation. When is a causal query determined by what we observe and assume?

Definition (Identifiability; Pearl 2000, Def. 3.2.2)

The causal effect of $\mathbf{X}$ on $\mathbf{Y}$ is *identifiable* from $\mathcal{G}$ if $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ can be computed uniquely from any positive distribution over the observed variables; that is, for every pair of models $\mathcal{M}_1, \mathcal{M}_2$ inducing $\mathcal{G}$,

$$P^{\mathcal{M}_1}(\mathbf{v}) = P^{\mathcal{M}_2}(\mathbf{v}) > 0 \implies P^{\mathcal{M}_1}(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = P^{\mathcal{M}_2}(\mathbf{y} \mid \mathrm{do}(\mathbf{x})).$$

Picture. The unobserved $\mathcal{M}$ induces $\mathcal{G}$ (known), $P(\mathbf{v})$ (observed) and $P(\mathbf{v} \mid \mathrm{do}(\mathbf{x}))$ (unobserved). The query is identifiable iff it is constant over all models inducing both $\mathcal{G}$ and $P(\mathbf{V})$.

Example 2.2 (A non-identifiable effect [Supplement])

$\mathcal{G}$: $X \to Y$, $X \leftrightarrow Y$, binary, shared $U \sim \mathrm{Bern}(1/2)$. $\mathcal{M}_1$: $X \leftarrow U$, $Y \leftarrow U$; $\mathcal{M}_2$: $X \leftarrow U$, $Y \leftarrow X$. Both give $P(X=Y=0) = P(X=Y=1) = 1/2$, but $P^{\mathcal{M}_1}(Y=1 \mid \mathrm{do}(X=1)) = 1/2 \neq 1 = P^{\mathcal{M}_2}(Y=1 \mid \mathrm{do}(X=1))$. (A graph permits, never requires, dependence.)

2.1.4.1 Identification in Markovian models

Theorem 2.2

Given the causal diagram $\mathcal{G}$ of any Markovian model (all relevant variables measured), $Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is identifiable for all disjoint $\mathbf{X}, \mathbf{Y}$, and

$$P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x})) = \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i),$$
$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{v}' \setminus \mathbf{y}} \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i).$$

Proof [Supplement]. By the truncated factorization,

$$P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x})) = \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i) \quad \text{evaluated at } \mathbf{X} = \mathbf{x},$$

each factor a conditional of $P(\mathbf{v})$, unique under positivity. Marginalizing over $\mathbf{V} \setminus (\mathbf{X} \cup \mathbf{Y})$ expresses $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ through $P(\mathbf{v})$ alone, so models agreeing on $\mathcal{G}$ and $P(\mathbf{v})$ agree on it. ■

2.1.4.2 Adjustment by direct parents (singleton)

Theorem 2.3

In a Markovian model where only a subset of variables is measured, $Q = P(\mathbf{y} \mid \mathrm{do}(x))$ is identifiable whenever X, $\mathbf{Y}$ and Pa_X are measured, and

$$P(\mathbf{y} \mid \mathrm{do}(x)) = \sum_{\mathbf{pa}_X} P(\mathbf{y} \mid x, \mathbf{pa}_X) P(\mathbf{pa}_X).$$

Proof [Supplement]. Sum the singleton reweighting identity

$$P(\mathbf{v}' \mid \mathrm{do}(x)) = P(\mathbf{v}'' \mid x, \mathbf{pa}_X) P(\mathbf{pa}_X)$$

over $\mathbf{v}'' \setminus \mathbf{y}$; unmeasured variables are summed out inside $P(\cdot \mid x, \mathbf{pa}_X)$. ■

2.1.4.3 Adjustment by direct parents for a set of treatments (optional)

For $\mathbf{X} = \{X_1, \ldots, X_k\}$ a parent of X_j may descend from an earlier X_i. Notation:

$Pa_{\mathbf{X}} = \bigcup_{X \in \mathbf{X}} Pa_X$	parents of $\mathbf{X}$ (may intersect $\mathbf{X}$),
$Pa^{-}_{\mathbf{W}} = Pa_{\mathbf{W}} \setminus \mathbf{W}$	parents of $\mathbf{W}$ excluding $\mathbf{W}$,
$\mathbf{X}_{<i} = \{X_1, \ldots, X_{i-1}\}$, $\mathbf{X}_{\le i}$	predecessors in a topological order,
$Pa'_{X_i} = Pa_{X_i} \setminus (\mathbf{X}_{<i} \cup Pa^{-}_{\mathbf{X}_{<i}})$	"new" parents of X_i.

Theorem 2.4

Let $\mathbf{X}$ be topologically ordered. If no element of Pa'_{X_j} is a descendant of X_i for every $i < j$, then

$$\prod_{X \in \mathbf{X}} P(x \mid \mathbf{pa}_X) = P(\mathbf{x} \mid \mathbf{pa}^{-}_{\mathbf{X}}),$$

hence

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{pa}^{-}_{\mathbf{X}}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{pa}^{-}_{\mathbf{X}}) P(\mathbf{pa}^{-}_{\mathbf{X}}).$$

Proof [Supplement]. Induction on i: $\prod_{X\in\mathbf{X}_{\le i}} P(x \mid \mathbf{pa}_X) = P(\mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le i}})$.

Base case. $P(x_1 \mid \mathbf{pa}_{X_1}) = P(x_1 \mid \mathbf{pa}_{X_1} \setminus \{X_1\}) = P(x_1 \mid \mathbf{pa}^{-}_{X_1}) = P(\mathbf{x}_{\le 1} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le 1}})$.

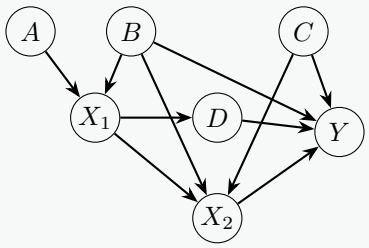
Step. Show $P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})P(x_i \mid \mathbf{pa}_{X_i}) = P(\mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le i}})$, using $Pa^{-}_{\mathbf{X}_{\le i}} = Pa^{-}_{\mathbf{X}_{<i}} \cup Pa'_{X_i}$:

$$
\begin{aligned}
&P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})P(x_i \mid \mathbf{pa}_{X_i}) \\
&= P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})P(x_i \mid \mathbf{pa}'_{X_i}, \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}}) && \text{Markov condition} \\
&= P(\mathbf{x}_{<i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})P(x_i \mid \mathbf{pa}'_{X_i}, \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})\frac{P(\mathbf{pa}'_{X_i} \mid \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})}{P(\mathbf{pa}'_{X_i} \mid \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})} && \text{multiply by 1} \\
&= \frac{P(\mathbf{pa}'_{X_i}, \mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})}{P(\mathbf{pa}'_{X_i} \mid \mathbf{x}_{<i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}})} && \text{chain rule, } \{X_i\} \cup \mathbf{X}_{<i} = \mathbf{X}_{\le i} \\
&= \frac{P(\mathbf{pa}'_{X_i}, \mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})}{P(\mathbf{pa}'_{X_i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{<i}})} && \mathbf{X}_{<i} \perp\!\!\!\perp Pa'_{X_i} \mid Pa^{-}_{\mathbf{X}_{<i}} \\
&= P(\mathbf{x}_{\le i} \mid \mathbf{pa}'_{X_i}, \mathbf{pa}^{-}_{\mathbf{X}_{<i}}) = P(\mathbf{x}_{\le i} \mid \mathbf{pa}^{-}_{\mathbf{X}_{\le i}}).
\end{aligned}
$$

Line 1: local Markov ($\mathbf{X}_{<i}$, $Pa^{-}_{\mathbf{X}_{<i}}$ are non-descendants of X_i). The key independence is d-separation: Pa'_{X_i} are non-descendants of every X_j, $j < i$, so a path from $\mathbf{X}_{<i}$ to Pa'_{X_i} either enters X_j through a parent (blocked by $Pa^{-}_{\mathbf{X}_{<i}}$) or leaves X_j along a directed edge and meets a collider that is not an ancestor of the conditioning set. ■

Example 2.3

$A \to X_1$, $B \to X_1$, $B \to X_2$, $B \to Y$, $C \to X_2$, $C \to Y$, $X_1 \to D$, $X_1 \to X_2$, $D \to Y$, $X_2 \to Y$.



$Pa_{X_1} = \{A, B\}$, $Pa_{X_2} = \{X_1, B, C\}$, $Pa^{-}_{\mathbf{X}} = \{A, B, C\}$, $Pa'_{X_2} = \{C\}$ (not a descendant of X_1). $P(y \mid \mathrm{do}(\mathbf{x})) = \sum P(\mathbf{v})/[P(x_1 \mid a, b)P(x_2 \mid x_1, b, c)]$, and

$$
\begin{aligned}
P(x_1 \mid a, b)P(x_2 \mid x_1, b, c) &= P(x_1 \mid a, b)P(x_2 \mid \underbrace{x_1}_{\mathbf{X}_{<2}}, \underbrace{a, b}_{Pa^{-}_{\mathbf{X}_{<2}}}, \underbrace{c}_{Pa'_{X_2}}) \\
&= P(x_1 \mid a, b)P(x_2 \mid x_1, a, b, c)\frac{P(c \mid x_1, a, b)}{P(c \mid x_1, a, b)} \\
&= \frac{P(x_1, x_2, c \mid a, b)}{P(c \mid x_1, a, b)} = \frac{P(x_1, x_2, c \mid a, b)}{P(c \mid a, b)} \\
&= P(x_1, x_2 \mid a, b, c) = P(\mathbf{x} \mid \mathbf{pa}^{-}_{\mathbf{X}}),
\end{aligned}
$$

so $P(y \mid \mathrm{do}(x_1, x_2)) = \sum_{a,b,c} P(y \mid x_1, x_2, a, b, c)P(a, b, c)$; D (a descendant of X_1) is correctly *not* adjusted for.

2.1.4.4 When a parent is latent

If Season is unobserved in the sprinkler model:

$$\begin{aligned} Q_2 &= \sum_{\text{sn,rn}} P(\text{sn})P(\text{rn} \mid \text{sn})P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn}) \\ &= \sum_{\text{sn,rn}} P(\text{rn}, \text{sn})P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn}) \\ &= \sum_{\text{rn}} P(\text{wt} \mid \text{Sp} = \text{on}, \text{rn})P(\text{rn}). \end{aligned}$$

Adjusting for Rain suffices; which sets work in general is the back-door criterion (Section 2.2).

2.1.5 Summary

- $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$ is defined through submodels.
- In Markovian models every effect is identifiable via the truncated factorization.
- Adjustment by direct parents: condition on Pa_X and average over their marginal.
- Identification asks whether $\mathcal{G}$ and $P(\mathbf{v})$ determine $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$.
- With latent variables it becomes non-trivial (back-door criterion and beyond).

2.2 Confounding and the back-door criterion

Motivation. How to identify an effect when some parents of the treatment are unobserved, and how to evaluate the adjustment formula from samples.

2.2.1 Recap: identifiable and non-identifiable effects

Motivation. Both sides of the identifiability definition, with concrete pairs of models.

2.2.1.1 An identifiable effect

$\mathcal{G}$: $Z \to X$, $Z \to Y$, $X \to Y$. Any compatible model gives

$$\begin{aligned} P(\mathbf{v}) &= P(z)P(x \mid z)P(y \mid x, z), \\ P(\mathbf{v}' \mid \text{do}(x)) &= P(z)P(y \mid x, z), \\ P(y \mid \text{do}(x)) &= \sum_z P(z)P(y \mid x, z). \end{aligned}$$

For two models

$$\mathcal{M}^{(j)} = \begin{cases} Z \leftarrow f_z^{(j)}(u_z) \\ X \leftarrow f_x^{(j)}(z, u_x) \\ Y \leftarrow f_y^{(j)}(x, z, u_y) \end{cases}$$

$$\mathcal{M}_x^{(j)} = \begin{cases} Z \leftarrow f_z^{(j)}(u_z) \\ X \leftarrow x \\ Y \leftarrow f_y^{(j)}(x, z, u_y) \end{cases}$$

$$j = 1, 2,$$

agreement on $\langle \mathcal{G}, P(\mathbf{v}) \rangle$ fixes $P(z)$ and $P(y \mid x, z)$, hence $P(\mathbf{v}' \mid \text{do}(x))$.

2.2.1.2 A non-identifiable effect

$\mathcal{G}$: $X \to Y$, $X \leftrightarrow Y$ (latent U_{xy}); U_y, U_{xy} independent fair coins, $X \leftarrow U_{xy}$:

$$\mathcal{M}^{(1)}: \; Y \leftarrow (X \oplus U_{xy}) \vee U_y, \qquad \mathcal{M}^{(2)}: \; Y \leftarrow U_y.$$

Observationally $X \oplus U_{xy} = 0$, so $Y = U_y$ in both. Under $\mathrm{do}(x)$:

U_y	U_{xy}	X	$Y^{(1)}$	$Y^{(2)}$	$P(\mathbf{u})$
0	0	x	x	0	1/4
0	1	x	$\neg x$	0	1/4
1	0	x	1	1	1/4
1	1	x	1	1	1/4

Y	$P^{(1)}(Y \mid \mathrm{do}(x))$	$P^{(2)}(Y \mid \mathrm{do}(x))$
0	1/4	1/2
1	3/4	1/2

($Y^{(1)} = 0$ needs $U_y = 0$, $x = U_{xy}$: probability 1/4.) Same $\mathcal{G}$ and $P(\mathbf{v})$, different effects: $P(\mathbf{v} \mid \mathrm{do}(x)) = \sum_{u_{xy}} P(y \mid x, u_{xy}) P(u_{xy})$ depends on the latent U_{xy}.

2.2.2 Confounding bias

Motivation. A common cause makes conditioning differ from intervening.

Exercise and cholesterol, both affected by age: pooled data show cholesterol *increasing* with exercise (older people exercise more), while within each age group it decreases. So

$$P(\text{cholesterol} \mid \text{exercise}) \neq P(\text{cholesterol} \mid \mathrm{do}(\text{exercise})),$$

the **confounding bias**.

Is confounding bias removable? If some parents of X are unobserved, parent adjustment (Section 2.1) fails; which measured sets $Z_1, \ldots, Z_k$ can replace them?

2.2.3 The back-door criterion

Motivation. A graphical condition under which a covariate set substitutes for the treatment's parents.

Definition (Back-door criterion; Pearl 2000, Def. 3.3.1)

A set $\mathbf{Z}$ satisfies the back-door criterion relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$ if

(i) no node in $\mathbf{Z}$ is a descendant of $\mathbf{X}$; and

(ii) $\mathbf{Z}$ blocks every path between $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$ that contains an arrow into X (a *back-door path*).

Theorem 2.5 (Back-door adjustment)

If $\mathbf{Z}$ satisfies the back-door criterion relative to $(\mathbf{X}, \mathbf{Y})$, the effect is identifiable and

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}).$$

Proof [Supplement]. We use Rules 2 and 3 of do-calculus (Chapter 3, proved from the truncated factorization alone, so no circularity). By the axioms,

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z})\, P(\mathbf{z} \mid \mathrm{do}(\mathbf{x})).$$

Step 1. $P(\mathbf{z} \mid \mathrm{do}(\mathbf{x})) = P(\mathbf{z})$. In $\mathcal{G}_{\overline{\mathbf{X}}}$ all edges at $\mathbf{X}$ point out, so a collider-free path from $\mathbf{X}$ to $\mathbf{Z}$ would be directed, making part of $\mathbf{Z}$ a descendant of $\mathbf{X}$, excluded by (i); other paths contain a collider and are blocked given $\emptyset$. So $(\mathbf{Z} \perp_d \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}}$; apply Rule 3.

Step 2. $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}) = P(\mathbf{y} \mid \mathbf{x}, \mathbf{z})$. In $\mathcal{G}_{\underline{\mathbf{X}}}$ every path between $\mathbf{X}$ and $\mathbf{Y}$ starts with an arrow into $\mathbf{X}$, i.e. is a back-door path, blocked by $\mathbf{Z}$ by (ii). So $(\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{Z})_{\mathcal{G}_{\underline{\mathbf{X}}}}$; apply Rule 2.

Step 3. Substitute both identities. ■

2.2.3.1 Back-door sets substitute for the parents: an example

In the sprinkler network, Rain is back-door admissible for (Sprinkler, Wet):

(i) Rain is not a descendant of Sprinkler;

(ii) Rain blocks the only back-door path Sprinkler ← Season → Rain → Wet. From parent adjustment,

$$\begin{aligned} P(\mathrm{wt} \mid \mathrm{do}(\mathrm{sp})) &= \sum_{\mathrm{sn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{sn}) P(\mathrm{sn}) \\ &= \sum_{\mathrm{sn},\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{sn}, \mathrm{rn}) P(\mathrm{rn} \mid \mathrm{sp}, \mathrm{sn}) P(\mathrm{sn}) \\ &= \sum_{\mathrm{sn},\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{rn}) P(\mathrm{rn} \mid \mathrm{sn}) P(\mathrm{sn}) \\ &= \sum_{\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{rn}) \sum_{\mathrm{sn}} P(\mathrm{rn}, \mathrm{sn}) \\ &= \sum_{\mathrm{rn}} P(\mathrm{wt} \mid \mathrm{sp}, \mathrm{rn}) P(\mathrm{rn}), \end{aligned}$$

the third equality by Wet ⫫ Season | Sprinkler, Rain and Rain ⫫ Sprinkler | Season (d-separation).

2.2.3.2 General proof: back-door adjustment = parent adjustment

With $Pa_{\mathbf{X}}^{-}$ the parents of $\mathbf{X}$ outside $\mathbf{X}$ and $\mathbf{Z}^{-} = \mathbf{Z} \setminus Pa_{\mathbf{X}}^{-}$:

$$\begin{aligned} \text{(i)} &\implies \mathbf{X} \perp\!\!\!\perp (\mathbf{Z} \setminus Pa_{\mathbf{X}}^{-}) \mid Pa_{\mathbf{X}}^{-} && \text{(local Markov)}, \\ \text{(ii)} &\implies \mathbf{Y} \perp\!\!\!\perp (Pa_{\mathbf{X}}^{-} \setminus \mathbf{Z}) \mid \mathbf{Z}, \mathbf{X} && \text{(back-door paths blocked)}, \end{aligned}$$

the first as $\mathbf{Z}$ are non-descendants of $\mathbf{X}$, the second as a path from a parent of $\mathbf{X}$ to $\mathbf{Y}$ avoiding $\mathbf{X}$ extends to a back-door path. Then

$$\begin{aligned} P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) &= \sum_{\mathbf{pa}_{\mathbf{X}}^{-}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{pa}_{\mathbf{X}}^{-}) P(\mathbf{pa}_{\mathbf{X}}^{-}) && \text{parent adjustment} \\ &= \sum_{\mathbf{z}^{-}, \mathbf{pa}_{\mathbf{X}}^{-}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{pa}_{\mathbf{X}}^{-}, \mathbf{z}^{-}) P(\mathbf{z}^{-} \mid \mathbf{x}, \mathbf{pa}_{\mathbf{X}}^{-}) P(\mathbf{pa}_{\mathbf{X}}^{-}) && \text{def. of cond. probability} \\ &= \sum_{\mathbf{z}^{-}, \mathbf{pa}_{\mathbf{X}}^{-}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}^{-} \mid \mathbf{pa}_{\mathbf{X}}^{-}) P(\mathbf{pa}_{\mathbf{X}}^{-}) && \text{(ii) and (i)} \\ &= \sum_{\mathbf{z}^{-}, \mathbf{pa}_{\mathbf{X}}^{-}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}^{-}, \mathbf{pa}_{\mathbf{X}}^{-}) \end{aligned}$$

$$= \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) \sum_{\mathbf{pa}_{\mathbf{X}}^- \setminus \mathbf{z}} P(\mathbf{z}^-, \mathbf{pa}_{\mathbf{X}}^-) \qquad \text{by algebra}$$
$$= \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}).$$

Line 3: $\{\mathbf{x}, \mathbf{pa}_{\mathbf{X}}^-, \mathbf{z}^-\} = \{\mathbf{x}, \mathbf{z}\} \cup (\mathbf{pa}_{\mathbf{X}}^- \setminus \mathbf{z})$, so (ii) drops $\mathbf{pa}_{\mathbf{X}}^- \setminus \mathbf{z}$;

(i) drops $\mathbf{x}$ from $P(\mathbf{z}^- \mid \cdot)$. *Back-door adjustment equals parent adjustment.* (The parent formula holds in the Markovian model including latents; the result uses only measured variables.)

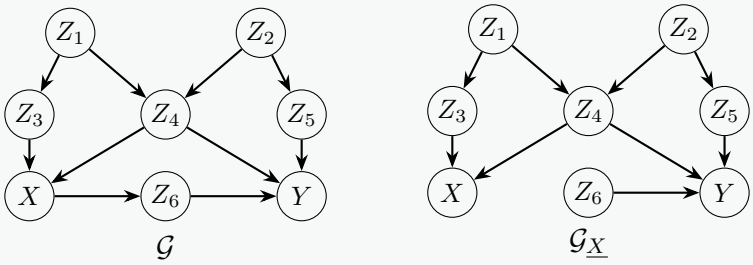
2.2.3.3 A graphical test

With $\mathcal{G}_{\underline{\mathbf{X}}}$ the graph without edges *out of* $\mathbf{X}$,

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) \text{ is identified by } \mathbf{Z} \quad \text{if (i) and } (\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{Z})_{\mathcal{G}_{\underline{\mathbf{X}}}}.$$

Example 2.4 (Pearl's seven-variable model)

$Z_1 \to Z_3 \to X$, $Z_1 \to Z_4$, $Z_2 \to Z_4$, $Z_2 \to Z_5 \to Y$, $Z_4 \to X$, $Z_4 \to Y$, $X \to Z_6 \to Y$.



$\mathcal{G}$ $\mathcal{G}_{\underline{X}}$

Back-door paths: $X \leftarrow Z_4 \to Y$, $X \leftarrow Z_4 \leftarrow Z_2 \to Z_5 \to Y$, $X \leftarrow Z_3 \leftarrow Z_1 \to Z_4 \to Y$, $X \leftarrow Z_3 \leftarrow Z_1 \to Z_4 \leftarrow Z_2 \to Z_5 \to Y$. Z_4 blocks the first three but opens the collider on the last, which Z_1 blocks. So $\{Z_1, Z_4\}$ is admissible:

$$P(y \mid \mathrm{do}(x)) = \sum_{z_1, z_4} P(y \mid x, z_1, z_4) P(z_1, z_4).$$

Also $\{Z_3, Z_4\}$, $\{Z_4, Z_2\}$, $\{Z_4, Z_5\}$; not $\{Z_4\}$ alone, never Z_6 (descendant of X).

Example 2.5 (More back-door examples [Supplement])

(a) $Z_2 \to Z_4$, $Z_2 \to Z_5$, $Z_4 \to X$, $Z_4 \to Y$, $X \leftrightarrow Z_4$, $X \to Z_6 \to Y$, $Z_5 \to Y$. Z_4 is needed for $X \leftarrow Z_4 \to Y$ but is a collider on $X \leftrightarrow Z_4 \leftarrow Z_2 \to Z_5 \to Y$, so add Z_2 or Z_5: $\mathbf{Z} \in \{\{Z_4, Z_2\}, \{Z_4, Z_5\}, \{Z_4, Z_2, Z_5\}\}$.
(b) $X \to Z$, $X \to Y$: no back-door path, Z a descendant, so $\mathbf{Z} = \emptyset$ and $P(y \mid \mathrm{do}(x)) = P(y \mid x)$.

2.2.4 Evaluating back-door adjustment: inverse probability weighting

Motivation. An estimator whose cost does not explode with the dimension of $\mathbf{Z}$.

For admissible $\mathbf{Z}$,

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}),$$
$$\mathbb{E}[Y \mid \mathrm{do}(\mathbf{x})] = \mathbb{E}_{\mathbf{Z}}[\mathbb{E}[Y \mid \mathbf{x}, \mathbf{Z}]].$$

Naively this needs many conditionals and a sum over $\mathbf{Z}$: about $O(\exp(|\mathbf{Z}|))$ time, with sparse cells.

2.2.4.1 Rewriting with the propensity score

$$P(y \mid \mathrm{do}(x)) = \sum_{\mathbf{z}} P(y \mid x, \mathbf{z}) P(\mathbf{z}) = \sum_{\mathbf{z}} \frac{P(y, x, \mathbf{z})}{P(x, \mathbf{z})} P(\mathbf{z}) = \sum_{\mathbf{z}} \frac{P(y, x, \mathbf{z})}{P(x \mid \mathbf{z})},$$

with the propensity fitted as $g(\mathbf{z}) \approx P(x \mid \mathbf{z})$.

2.2.4.2 Sample version

Replacing $P(y, x, \mathbf{z})$ by empirical frequencies,

$$\begin{aligned} P(y \mid \mathrm{do}(x)) &\approx \sum_{\mathbf{z}} \frac{\frac{1}{N} \sum_{i=1}^{N} \mathbb{1}(y_i = y, x_i = x, \mathbf{z}_i = \mathbf{z})}{g(\mathbf{z})} \\ &= \frac{1}{N} \sum_{i=1}^{N} \sum_{\mathbf{z}} \frac{\mathbb{1}(y_i = y, x_i = x, \mathbf{z}_i = \mathbf{z})}{g(\mathbf{z})} \\ &= \frac{1}{N} \sum_{i=1}^{N} \frac{\mathbb{1}(y_i = y, x_i = x)}{g(\mathbf{z}_i)}, \end{aligned}$$

since only $\mathbf{z} = \mathbf{z}_i$ contributes; cost $O(N)$, not $O(|\mathfrak{X}_{\mathbf{Z}}|)$. Similarly,

$$\begin{aligned} \mathbb{E}[Y \mid \mathrm{do}(x)] &= \sum_{\mathbf{z}, y} y \, \frac{P(y, x, \mathbf{z})}{P(x \mid \mathbf{z})} \\ &\approx \frac{1}{N} \sum_{i=1}^{N} \sum_{\mathbf{z}, y} \frac{y \mathbb{1}(y_i = y, x_i = x, \mathbf{z}_i = \mathbf{z})}{g(\mathbf{z})} \\ &= \frac{1}{N} \sum_{i=1}^{N} \frac{y_i \mathbb{1}(x_i = x)}{g(\mathbf{z}_i)}. \end{aligned}$$

2.2.4.3 Average treatment effect

Binary X, $g(\mathbf{z}) = P(X = 1 \mid \mathbf{z})$, back-door criterion holding:

$$\begin{aligned} \widehat{\mathrm{ATE}} &= \hat{\mathbb{E}}[Y \mid \mathrm{do}(X = 1)] - \hat{\mathbb{E}}[Y \mid \mathrm{do}(X = 0)] \\ &= \frac{1}{N} \sum_{i=1}^{N} y_i \left(\frac{\mathbb{1}(x_i = 1)}{g(\mathbf{z}_i)} - \frac{\mathbb{1}(x_i = 0)}{1 - g(\mathbf{z}_i)} \right) \\ &= \frac{1}{N} \sum_{i=1}^{N} \left(\frac{y_i x_i}{g(\mathbf{z}_i)} - \frac{y_i (1 - x_i)}{1 - g(\mathbf{z}_i)} \right). \end{aligned}$$

(Elsewhere covariates are often X or L, treatment T, W or A.)

2.2.4.4 Interpretation: pseudo-causal samples

IPW builds a pseudo-population mimicking a randomized experiment: drop samples with $x_i \neq x$ and replicate each remaining one $1/g(\mathbf{z}_i)$ times (e.g. $P(x_1 \mid \mathbf{z}_1) = 1/3$: three times; $P(x_3 \mid \mathbf{z}_3) = 1/2$: twice). This removes the dependence between X and $\mathbf{Z}$.

2.2.5 Summary

- A common cause of treatment and outcome gives $P(y \mid x) \neq P(y \mid \mathrm{do}(x))$.

- The back-door criterion tests admissibility of $\mathbf{Z}$ graphically.
- Back-door adjustment: $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$.
- Any back-door admissible set can substitute for the parents.
- IPW evaluates it from samples without summing over high-dimensional $\mathbf{Z}$.

2.3 The adjustment criterion

Motivation. Exactly which sets $\mathbf{Z}$ make $\sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$ equal $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$, and how to test, construct and enumerate them.

2.3.1 Back-door criterion: recap and two refinements

Motivation. Back-door adjustment resolves Simpson's paradox and extends to conditional effects.

Theorem 2.6 (Back-door adjustment; Pearl 2000, Thm. 3.3.2)

If $\mathbf{Z}$ satisfies (i) no node in $\mathbf{Z}$ is a descendant of $\mathbf{X}$, and (ii) $\mathbf{Z}$ blocks every path between $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$ containing an arrow into X, then $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$.

Proof [Supplement]. The proof of Section 2.2 (Rule 3 for $P(\mathbf{z} \mid \mathrm{do}(\mathbf{x}))$, Rule 2 for $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z})$) applies verbatim. ■

2.3.1.1 Back to Simpson's paradox

The drug/sex/recovery tables of Chapter 1 under the three stories:

	$F \to X,\ F \to Y,\ X \to Y$	$X \to F \to Y,\ X \to Y$	$X \leftarrow U_1 \to F \leftarrow U_2 \to Y,\ X \to Y$
$\emptyset$	not admissible	admissible	admissible
$\{F\}$	admissible	not admissible ($F \in De(X)$)	not admissible (opens collider)
Use	F-specific tables	aggregated table	aggregated table

The physician's dilemma becomes a mechanical graph check.

2.3.1.2 Conditional back-door

Theorem 2.7 (Conditional back-door criterion)

The $\mathbf{w}$-specific effect $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$ is identified whenever we can measure a set $\mathbf{Z}$ such that $\mathbf{W} \cup \mathbf{Z}$ satisfies the back-door criterion. Moreover,

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w}) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}, \mathbf{w}) P(\mathbf{z} \mid \mathbf{w}).$$

Proof [Supplement]. $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w}) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) P(\mathbf{z} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$.

Step 1. All paths in $\mathcal{G}_{\underline{\mathbf{X}}}$ are back-door paths, blocked by $\mathbf{W} \cup \mathbf{Z}$: $(\mathbf{Y} \perp_d \mathbf{X} \mid \mathbf{Z}, \mathbf{W})_{\mathcal{G}_{\underline{\mathbf{X}}}}$, so Rule 2 gives $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}, \mathbf{w})$.

Step 2. $\mathbf{W}$ has no descendant of $\mathbf{X}$, so $\mathbf{X}(\mathbf{W}) = \mathbf{X}$ and Rule 3 uses $\mathcal{G}_{\overline{\mathbf{X}}}$. A path from $\mathbf{X}$ to $\mathbf{Z}$ there leaves $\mathbf{X}$ outward; collider-free, it would make $\mathbf{Z}$ contain a descendant of $\mathbf{X}$; otherwise its first

collider C and all descendants of C descend from $\mathbf{X}$, so are not in $\mathbf{W}$, and the path is blocked. Thus $(\mathbf{Z} \perp_d \mathbf{X} \mid \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$ and $P(\mathbf{z} \mid \mathrm{do}(\mathbf{x}), \mathbf{w}) = P(\mathbf{z} \mid \mathbf{w})$.

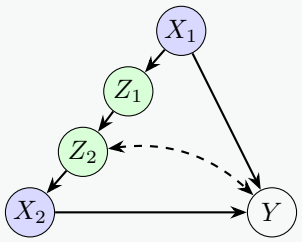
Step 3. Substitute. ■

2.3.2 Tightness of the back-door criterion

Motivation. The back-door criterion is sufficient but not necessary.

Example 2.6

$\mathbf{X} = \{X_1, X_2\}$ with $X_1 \to Z_1 \to Z_2 \to X_2$, $X_1 \to Y$, $X_2 \to Y$, $Z_2 \leftrightarrow Y$.



All other variables descend from X_1, so no back-door set exists; yet with $\mathbf{Z} = \{Z_1, Z_2\}$,

$$P(y \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(y \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$$

(Z_2 blocks $X_2 \leftarrow Z_2 \leftrightarrow Y$; Z_1 blocks the opened $X_1 \to Z_1 \to Z_2 \leftrightarrow Y$).

Condition (i) protects causal paths, but some descendants of $\mathbf{X}$ (e.g. ones that are also ancestors of $\mathbf{X}$ off the causal paths) may, or must, be used. *Which variables must really be avoided?*

2.3.3 The adjustment criterion

Motivation. Necessary and sufficient conditions for valid adjustment sets.

Back-door ⇒ adjustment formula; we seek "???" with

$$\mathbf{Z} \text{ is ??? w.r.t. } \langle \mathbf{X}, \mathbf{Y} \rangle \iff P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z}).$$

2.3.3.1 Proper causal paths

Definition (Proper causal path; Shpitser et al. 2010)

A path from $X \in \mathbf{X}$ to $Y \in \mathbf{Y}$ is *causal* if it is directed, and *proper* if it contains a member of $\mathbf{X}$ only at its starting point.

Example 2.7

$X_1 \to W_1 \to X_2 \to W_2 \to Y$, $X_1 \to W_3 \to Y$, $X_1 \leftrightarrow Y$. Proper causal: $X_1 \to W_3 \to Y$, $X_2 \to W_2 \to Y$; not proper: $X_1 \to W_1 \to X_2 \to W_2 \to Y$.

2.3.3.2 Disturbing a proper causal path

Adjusting for a variable on a proper causal path (other than $\mathbf{X}$) or a descendant of one *disturbs* it.

Example 2.8

$X \to W \to Y$, $W \to Z$: $P(y \mid \mathrm{do}(x)) = P(y \mid x)$. Adjusting for Z:

$$\begin{aligned}\sum_z P(y \mid x, z)P(z) &= \sum_{z,w} P(y \mid x, z, w)P(w \mid x, z)P(z) \\ &= \sum_w P(y \mid w) \sum_z P(w \mid x, z)P(z), \\ P(y \mid \mathrm{do}(x)) = P(y \mid x) &= \sum_{w,z} P(y \mid x, z, w)P(w \mid x, z)P(z \mid x) \\ &= \sum_w P(y \mid w) \sum_z P(w \mid x, z)P(z \mid x).\end{aligned}$$

They differ whenever $P(z) \neq P(z \mid x)$ (generic, as Z depends on X via W); likewise for adjusting for W.

2.3.3.3 The criterion

Theorem 2.8 (Adjustment criterion; Shpitser, VanderWeele and Robins 2010)

$\mathbf{Z}$ satisfies the adjustment criterion relative to $(\mathbf{X}, \mathbf{Y})$ in $\mathcal{G}$ iff

(i) $\mathbf{Z}$ does not disturb any proper causal path: no element of $\mathbf{Z}$ is a descendant, in $\mathcal{G}_{\overline{\mathbf{X}}}$, of any $W \notin \mathbf{X}$ lying on a proper causal path from $\mathbf{X}$ to $\mathbf{Y}$; and

(ii) all proper non-causal paths in $\mathcal{G}$ from $\mathbf{X}$ to $\mathbf{Y}$ are blocked by $\mathbf{Z}$.

$\mathbf{Z}$ satisfies the adjustment criterion *if and only if* $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z})P(\mathbf{z})$ for every model compatible with $\mathcal{G}$.

Proof [Supplement]. *Sufficiency when* $\mathbf{Z}$ *has no descendant of* $\mathbf{X}$. (i) is automatic. A non-causal path either starts into $\mathbf{X}$ (back-door) or starts out of $\mathbf{X}$ and is not directed, hence has a collider descending from $\mathbf{X}$, outside $\mathbf{Z}$ with no descendant in $\mathbf{Z}$: blocked. So (ii) is the back-door criterion and the back-door theorem applies.

General case. Sufficiency and necessity are Shpitser, VanderWeele and Robins (2010, Theorem 3), via the proper back-door graph; necessity builds, for each violating $\mathbf{Z}$, two models agreeing on $P(\mathbf{v})$ but not on the formula. Not reproduced here. ■

2.3.3.4 Implementation idea

Input: $\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathcal{G}$. **Output:** whether $\mathbf{Z}$ is adjustment-admissible.

Step 1 (proper causal paths).

$$\mathrm{PCP} = (De(\mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}} \setminus \mathbf{X}) \cap An(\mathbf{Y})_{\mathcal{G}_{\underline{\mathbf{X}}}},$$

descendants of $\mathbf{X}$ reaching $\mathbf{Y}$ without re-entering $\mathbf{X}$. (Superset alternative: $\mathbf{W} = De(\mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}} \cap An(\mathbf{Y})_{\mathcal{G}_{\overline{\mathbf{X}}}}$, $\mathcal{H} = \mathcal{G}[\mathbf{W}]$ without bidirected edges; ancestors in $\mathcal{G}_{\underline{\mathbf{X}}}$ drop vertices reaching $\mathbf{Y}$ only through another treatment.)

Step 2 (forbidden variables). $\mathcal{F} = De(\mathrm{PCP})_{\mathcal{G}_{\overline{\mathbf{X}}}}$. *Check (i):* $\mathbf{Z} \cap \mathcal{F} \neq \emptyset \Rightarrow$ not admissible.

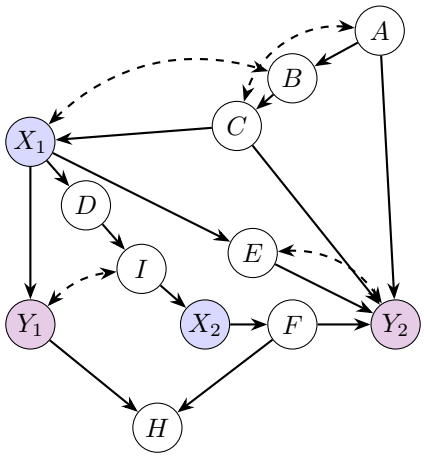
Step 3 (proper non-causal paths). Remove the first edge of every proper causal path,

$$\mathcal{G}' = \mathcal{G} \setminus \{X \to W : X \in \mathbf{X},\ W \in \mathrm{PCP}\}.$$

Check (ii): admissible iff $(\mathbf{X} \perp_d \mathbf{Y} \mid \mathbf{Z})_{\mathcal{G}'}$.

Each step is a graph search: polynomial.

2.3.3.5 An example



$\mathcal{G}$: $A \to B \to C \to X_1$, $C \to Y_2$, $A \to Y_2$, $X_1 \to D \to I \to X_2$, $X_1 \to E \to Y_2$, $X_1 \to Y_1$, $X_2 \to F \to Y_2$, $Y_1 \to H \leftarrow F$; $X_1 \leftrightarrow B$, $A \leftrightarrow C$, $Y_1 \leftrightarrow I$, $E \leftrightarrow Y_2$; $\mathbf{X} = \{X_1, X_2\}$, $\mathbf{Y} = \{Y_1, Y_2\}$.

(i) Forbidden set. Proper causal paths $X_1 \to Y_1$, $X_1 \to E \to Y_2$, $X_2 \to F \to Y_2$ ($X_1 \to D \to I \to X_2 \to \cdots$ is not proper); descendants of $\{Y_1, E, Y_2, F\}$ are $\{Y_1, E, Y_2, F, H\}$, so $\mathcal{F} = \{E, F, H\}$ besides $\mathbf{Y}$.

(ii) Evaluating $\mathbf{Z} = \{C, D, I\}$. $\mathbf{Z} \cap \mathcal{F} = \emptyset$. Cutting $X_1 \to Y_1$, $X_1 \to E$, $X_2 \to F$ gives $\mathcal{G}'$, where $X_1 \leftrightarrow B \leftarrow A \to Y_2$ is open (B a collider with descendant $C \in \mathbf{Z}$, A an unconditioned non-collider). So $(\mathbf{X} \not\perp_d \mathbf{Y} \mid C, D, I)_{\mathcal{G}'}$: *not* admissible. (Non-causal paths in $\mathcal{G}'$ are proper non-causal paths in $\mathcal{G}$.)

2.3.3.6 Explicit construction of an admissible set

Lemma 2.9 (Explicit admissible set)

Let $\mathcal{F}$ be the variables forbidden by condition (i) and $An^-(\mathbf{W}) = An(\mathbf{W}) \setminus \mathbf{W}$. If any admissible set exists, then

$$\mathbf{Z}_0 = An^-(\mathbf{X} \cup \mathbf{Y}) \setminus \mathcal{F}$$

is admissible. (Ancestors should be taken in $\mathcal{G}'$, but the result is the same in $\mathcal{G}$.)

Proof [Supplement]. By the constructive back-door criterion (van der Zander, Liśkiewicz and Textor 2014), $\mathbf{Z}$ is an adjustment set iff $\mathbf{Z} \cap \mathcal{F} = \emptyset$ and $\mathbf{Z}$ d-separates $\mathbf{X}, \mathbf{Y}$ in the proper back-door graph $\mathcal{G}'$. If an admissible $\mathbf{Z}$ exists, the inclusion–restriction lemma in $\mathcal{G}'$ with $\mathbf{I} = \emptyset$, $\mathbf{R} = \mathbf{V} \setminus (\mathcal{F} \cup \mathbf{X} \cup \mathbf{Y})$ shows that $An(\mathbf{X} \cup \mathbf{Y}) \cap \mathbf{R} = An^-(\mathbf{X} \cup \mathbf{Y}) \setminus \mathcal{F}$ also separates them in $\mathcal{G}'$ and avoids $\mathcal{F}$. ■

Here $An^-(\mathbf{X} \cup \mathbf{Y}) = \{A, B, C, D, E, F, I\}$, $\mathbf{Z}_0 = \{A, B, C, D, I\}$: A blocks $X_1 \leftrightarrow B \leftarrow A \to Y_2$ and $X_1 \leftrightarrow B \leftarrow A \leftrightarrow C \to Y_2$; C blocks $X_1 \leftarrow C \to Y_2$; D blocks $X_1 \to D \to I \leftrightarrow Y_1$ (collider I opened); I blocks $X_2 \leftarrow I \leftrightarrow Y_1$. So $\mathbf{Z}_0$ is admissible, although D, I descend from X_1 (never back-door admissible).

Testing existence. If $\mathbf{Z}_0$ is not admissible, none is: a *one-shot* test.

2.3.3.7 Recap

- The adjustment criterion characterizes all valid adjustment sets.
- Condition (i): $\mathbf{Z} \cap \mathcal{F} = \emptyset$.
- Condition (ii): $\mathbf{Z}$ blocks all proper non-causal paths.
- Testing and construction are polynomial.

2.3.4 Listing all admissible sets (optional)

Motivation. Enumerate all admissible sets with polynomial delay, to choose by cost, variance or other preferences.

2.3.4.1 Motivation

Choice may depend on availability, cost, variance, privacy or fairness. Example: $A \to X$, $A \to B \to Y$, $X \to C \to Y$: $\{A, B\}$, $\{A\}$, $\{B\}$ are admissible; which if A is a sensitive "genetic condition" and B is "headache"?

2.3.4.2 Challenge: output size

The number of solutions can be exponential: with k parallel paths $X \to W_{i,1} \to W_{i,2} \to Y$ there are 3^k separators. Total running time is the wrong yardstick.

Definition (Polynomial delay)

A listing algorithm runs with *polynomial delay* if there is a constant d such that it spends at most $O(n^d)$ time (1) between the start and either producing the first output or halting with no output, and (2) between producing one output and either producing the next or halting.

2.3.4.3 Theoretical tool: bounded separators

Lemma 2.10 (Inclusion and restriction)

Given $\mathcal{G}$ and disjoint $\mathbf{X}, \mathbf{Y}$, and $\mathbf{I} \subseteq \mathbf{R} \subseteq \mathbf{V}$: if any set $\mathbf{Z}$ with $\mathbf{I} \subseteq \mathbf{Z} \subseteq \mathbf{R}$ separates $\mathbf{X}$ and $\mathbf{Y}$ in $\mathcal{G}$, then

$$\mathbf{Z}_0 = An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{I}) \cap \mathbf{R}$$

also separates them.

Proof [Supplement]. Moralization: $\mathbf{S}$ d-separates $\mathbf{X}, \mathbf{Y}$ in $\mathcal{G}$ iff it separates them in $(\mathcal{G}_A)^m$, $A = An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{S})$; for ancestral $A \subseteq A'$, $(\mathcal{G}_A)^m$ is the subgraph of $(\mathcal{G}_{A'})^m$ induced by A (parents of A lie in A).

Let $A = An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{I})$. Then $\mathbf{I} \subseteq \mathbf{Z}_0 \subseteq A$, so $An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{Z}_0) = A$. If $\mathbf{Z}_0$ fails, some path π in $(\mathcal{G}_A)^m$ from $\mathbf{X}$ to $\mathbf{Y}$ has inner nodes in $A \setminus \mathbf{R}$, hence outside $\mathbf{Z}$. With $A' = An(\mathbf{X} \cup \mathbf{Y} \cup \mathbf{Z}) \supseteq A$, π is a path in $(\mathcal{G}_{A'})^m$ avoiding $\mathbf{Z}$, contradicting that $\mathbf{Z}$ separates. ■

$\mathbf{I}$: "include", $\mathbf{R}$: "restrict to". Testing $\mathbf{Z}_0$ in $O(n+m)$ decides existence: `exist-sep`$(\mathcal{G}, \mathbf{X}, \mathbf{Y}, \mathbf{I}, \mathbf{R})$. With $\mathbf{I} = \emptyset$, $\mathbf{R} = \mathbf{V} \setminus \mathcal{F}^+$ this is the explicit construction above.

2.3.4.4 Algorithm

`list-seps`$(\mathcal{G}, \mathbf{X}, \mathbf{Y}, \mathbf{I}, \mathbf{R})$ outputs all separators with $\mathbf{I} \subseteq \mathbf{Z} \subseteq \mathbf{R}$. With $\mathcal{F}^+ = \mathbf{X} \cup \mathbf{Y} \cup \mathcal{F}$, start from $\mathbf{I} = \emptyset$, $\mathbf{R} = \mathbf{V} \setminus \mathcal{F}^+$ in $\mathcal{G}'$; every output then satisfies (ii) and (i).

Algorithm 1 `list-seps`$(\mathcal{G}, \mathbf{X}, \mathbf{Y}, \mathbf{I}, \mathbf{R})$

```
if exist-sep(G, X, Y, I, R) then                          ▷ O(n + m)
    if I = R then output I
    else
        pick W ∈ R \ I
        list-seps(G, X, Y, I ∪ {W}, R)                    ▷ include W
        list-seps(G, X, Y, I, R \ {W})                    ▷ exclude W
```

Example 2.9 (Running example)

$A \to X$, $A \to B \to Y$, $D \to X$, $D \leftrightarrow Y$, $X \to C \to Y$.

1. $\mathcal{F}^+ = \{C, X, Y\}$.
2. $\mathbf{I} = \emptyset$, $\mathbf{R} = \{A, B, D\}$; remove $X \to C$.
3. Branch on D: excluding D is pruned ($X \leftarrow D \leftrightarrow Y$ unblockable). Including D, branch on A, then B: leaves $\{A, B, D\}$, $\{A, D\}$, $\{B, D\}$ (outputs) and $\{D\}$ (pruned: $X \leftarrow A \to B \to Y$ open).

The admissible sets are $\{A, B, D\}, \{A, D\}, \{B, D\}$.

2.3.4.5 Analysis

- A node is expanded only if `exist-sep` holds, so every expanded node has an output leaf below it.
- Each reached leaf outputs $\mathbf{Z} = \mathbf{I} = \mathbf{R}$, satisfying (ii) and, as $\mathbf{R}$ excludes $\mathcal{F}^+$, (i).
- Depth $\leq k = n - |\mathcal{F}^+| < n$, so $O(n)$ calls between consecutive outputs.
- Each call costs $O(n + m)$, so the delay is $O(n(n + m))$.

Total time may be exponential (unavoidable), but outputs come with polynomial delay, and the algorithm halts in polynomial time if there is none.

2.3.5 Connection with conditional ignorability

Motivation. The graphical criterion justifies the key potential-outcome assumption.

The PO framework assumes *conditional ignorability*,

$$\mathbf{Y}_{\mathbf{x}} \perp\!\!\!\perp \mathbf{X} \mid \mathbf{Z},$$

which gives $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}) = P(\mathbf{y} \mid \mathbf{x}, \mathbf{z})$ and the adjustment formula. The adjustment criterion *justifies* it: it says when such a $\mathbf{Z}$ exists given a model of reality.

2.3.6 Summary

- Adjustment is the most popular identification technique in the data sciences.

- The back-door/adjustment criterion gives a principled validity condition.
- Admissible sets can be found systematically and efficiently from the graph.
- IPW and variants then estimate the effect from finite samples.